ADSP 31014 Statistical Models for Data Science
Homework 2

(10 points) Question 1a
Intercept Only: In our lecture, for linear regression model with only the intercept parameter

$$y = \beta_0 + \epsilon$$

the optimal intercept parameter $\beta_0^* = \bar{y}_n$ is simply the sample mean of the observed response values. Let's complete the following R (or equivalent Python) code to verify.

```
y <- c(2, 5, 4.3, 4.2, 8) # Observed data

compute.mse <- Vectorize(function(b0) {
  return(mean((y - b0) ** 2))
})

b0 <- seq(0, 10, length.out = 10000 + 1) # Try various b0's
mse <- compute.mse(b0)
plot(...) # Plot b0 vs mse
```

Show the plot. It should have a quadratic parabola shape. What is the minimum MSE value? Which value of `b0` attains this minimum? (Hint: Use R's `which.min()` function.) Does this optimal `b0` value equal `mean(y)`?

(5 points) Question 1b
Use R's `lm()` function (or Python's `statsmodels` equivalence) to fit a linear regression model with only intercept to the data in Question 1a. Show the fitted model's `summary()`. (Hint: Do some research on how to write code for such model formula.)

(5 points) Question 2a
MAE: In our lecture, someone asked what if we fit linear regression model by minimizing the mean absolute error MAE objective function instead of MSE? Let's complete the following R (or equivalent Python) code to find out.

```
y <- c(2, 5, 4.3, 4.2, 8) # Observed data

compute.mae <- Vectorize(function(b0) {
  return(mean(abs(y - b0)))
})

b0 <- seq(0, 10, length.out = 10000 + 1) # Try various b0's
mae <- compute.mae(b0)
plot(...) # Plot b0 vs mae
```

Show the plot. How would you describe its shape? (Hint: Where are the y values located? What happens in-between them? It is not a quadratic parabola.)

(5 points) Question 2b

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Continuing from Question 2a, what is the minimum MAE value? Which value of `b0` attains this minimum? What is the relationship between this optimal β_0^* and the observed data $y_1, \ldots, y_n$? (Hint: It is not the sample mean but something similar.)

(5 points) Question 3
Ask some GenAI tool (e.g. ChatGPT, Gemini, Claude) to compare and contrast the choice of using MSE vs MAE for a regression problem. What does GenAI say? In what scenarios would one loss function be preferred over the other?

(30 points) Question 4a
Chicago Taxi Cost: In our lecture, we predict Chicago taxi trip duration based on other explanatory variables such as fare and tips %. Now let's consider another use case:

> "A rideshare company would like to enter the Chicago market with a competitive pricing strategy. To this end, they want to understand how Chicago taxi trip cost is related to other factors of the trip (that would be known before the trip begins)."

Let's use the `chicago_taxi_trip_sample.csv` data set.

- Fit a linear regression model with `trip_total` as the response variable and `trip_miles` as the explanatory variable. (Strictly speaking, exact trip distance is not known before a trip begins. In practice, we would estimate this based on pick-up / drop-off locations etc. For simplicity, here we will allow ourselves to use trip distance as-is.)
- What are the fitted intercept β_0^* and slope β_1^* parameters? Write down the fitted model equation in the form $y = \beta_0^* + \beta_1^* x$.
- Compute MSE of the fitted model on the observed data.
- What is the R^2 value for the fitted model?
- Compute sample correlation $\mathrm{Corr}(Y, \hat{Y})$ between the observed responses y's and the predictions $\hat{y}$'s. Does its squared value equal R^2?
- What is the predicted cost for a 10 miles trip?
- If a linear regression model has an intercept term β_0, then the sum of the residuals is always 0. (In other words, the model does not overestimate or underestimate on average.) Verify this fact numerically.

(10 points) Question 4b
In Question 4a scenario, what if the explanatory variable is in unit of kilometer instead of mile? Fit a linear regression model with `trip_total` as the response variable and `trip_km` as the explanatory variable.

```
data$trip_km <- data$trip_miles * 1.609
model <- lm(..., data = data)
```

- What are the fitted intercept β_0^* and slope β_1^* parameters? How are these related to the parameters in Question 3a? (Hint: It is an affine transformation.)
- What is the predicted cost for a 16.09 km trip? How is this related to the prediction in Question 3a?

(20 points) Question 4c
Continuing from Question 4a, let's include additional explanatory variable(s) besides `trip_miles`.

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- Fit a linear regression model with trip_miles and tolls as explanatory variables. What is the R^2 value for this fitted model?
- Fit a linear regression model with trip_miles and extras as explanatory variables. What is the R^2 value for this fitted model?
- Based on goodness-of-fit via R^2, which one of tolls vs extras is a more useful explanatory variable?
- Fit a linear regression model with trip_miles, tolls , and extras as explanatory variables. What is the R^2 value for this fitted model? Is this R^2 value even higher than the previous models?

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